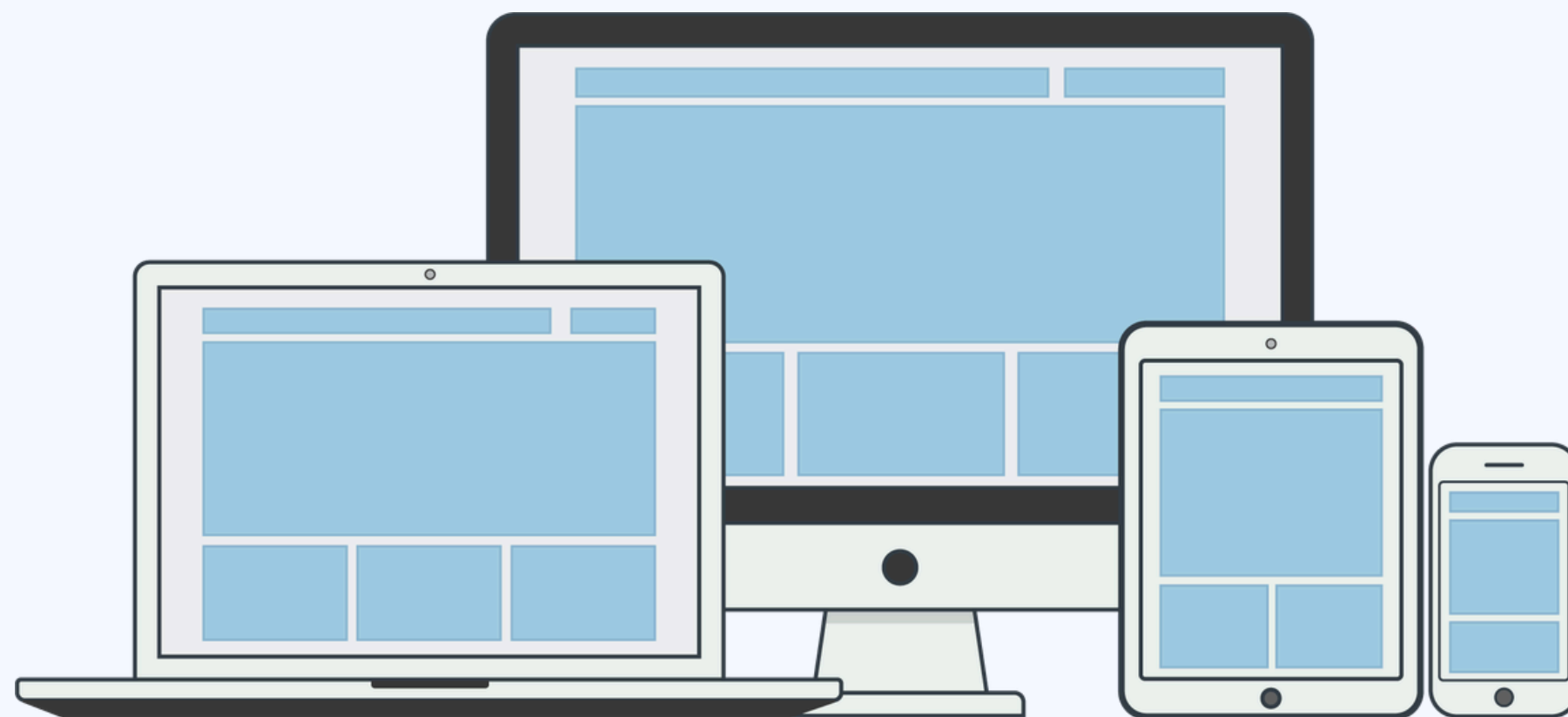


INTRODUCTION TO RELATIONAL DATABASES WITH POSTGRESQL



Course outcome

01 Get introduced to Databases

02 Understand Shema and Relational models

03 Understand Normalization in Database and its relevant

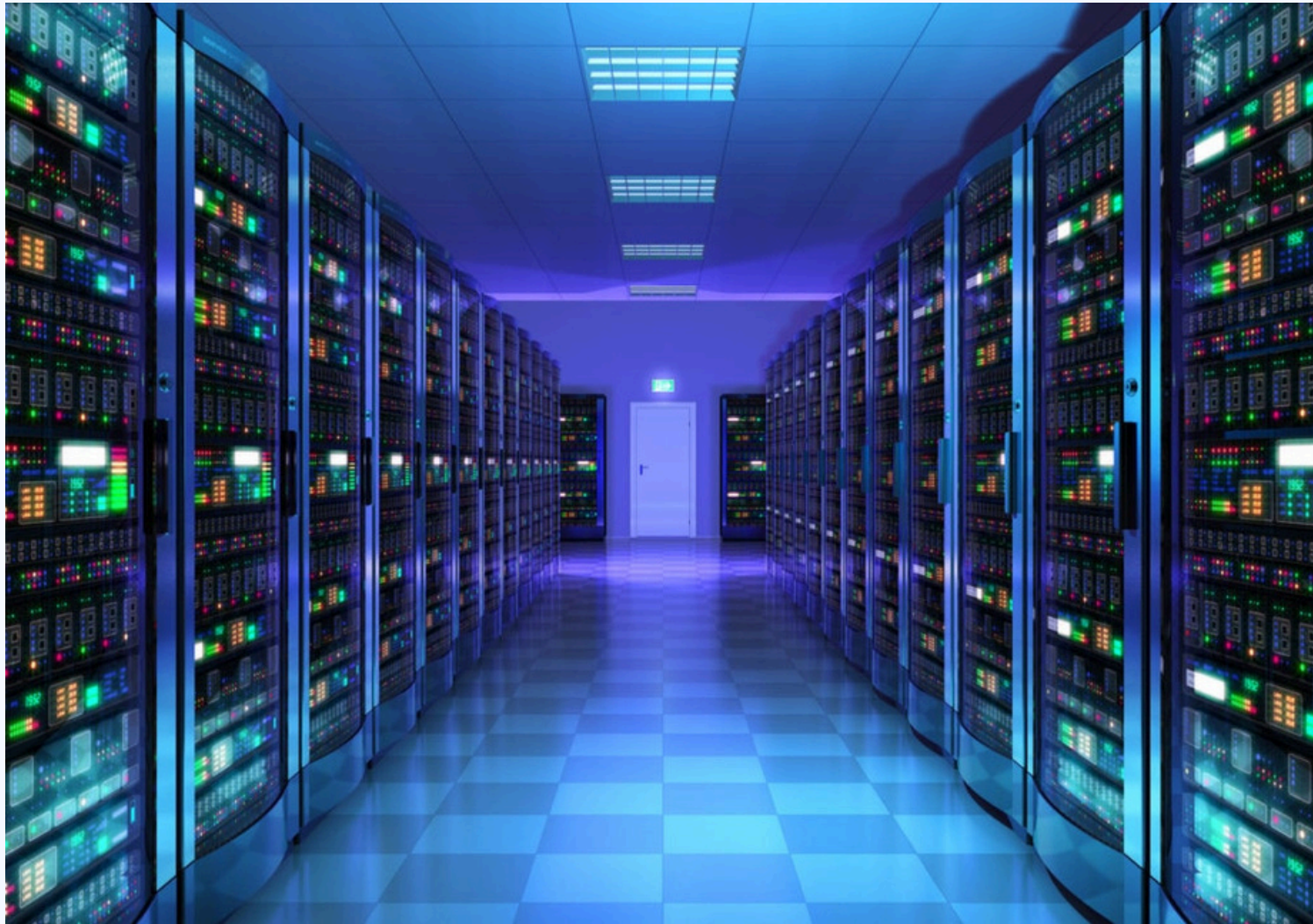
04 Know the various data types and constraints

05 Understanding data administration

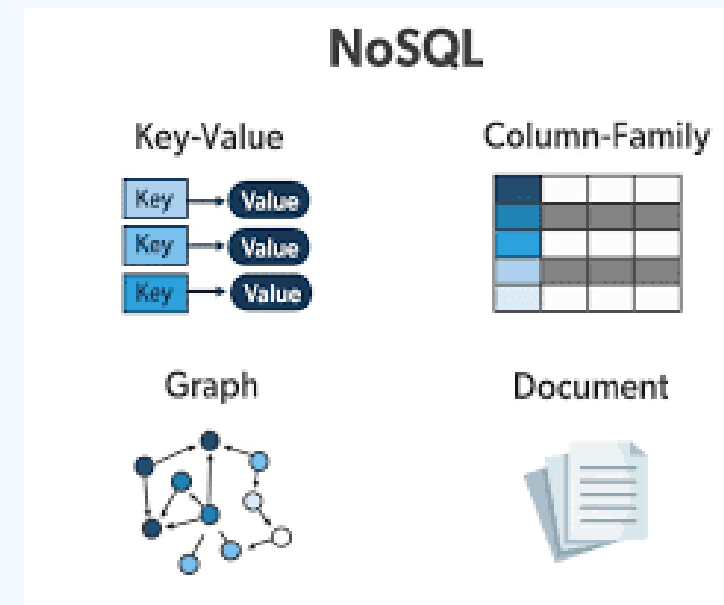
06 Get introduced to DBMS

07 Get introduced to PostgreSQL

Introduction to database



Database is a collection of data, and are designed to store, manage, and manipulate large volume of data. Databases come in two main forms known as:

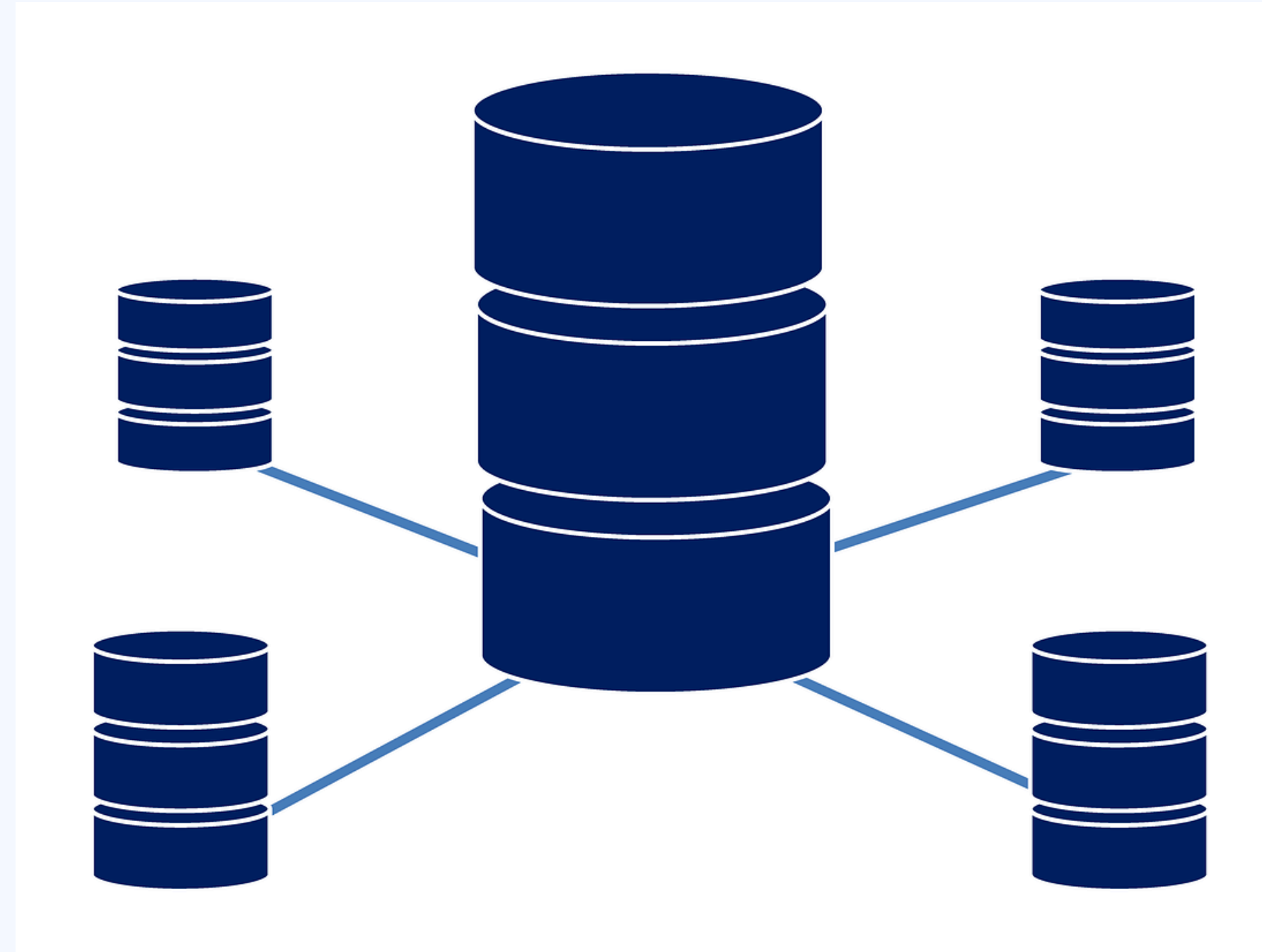


```
select * from sys.allocation_units
```

	allocation_unit_id	type	type_desc	container_id	data_space_id	total_pages	used_pages	data_pages
82	844424932884480	1	IN_ROW_DATA	844424932884480	1	0	0	0
83	844424933408768	1	IN_ROW_DATA	844424933408768	1	2	2	1
84	844424935768064	1	IN_ROW_DATA	844424935768064	1	0	0	0
85	1125899909070...	1	IN_ROW_DATA	1125899909070...	1	17	10	8
86	7177611906514...	2	LOB_DATA	281474980642816	1	12	6	0
87	7205759403799...	1	IN_ROW_DATA	281474980511744	1	2	2	1
88	7205759403805...	1	IN_ROW_DATA	56294995722400	1	2	2	1
89	7205759403819...	1	IN_ROW_DATA	281474983067648	1	2	2	1
90	7205759403825...	1	IN_ROW_DATA	562949959778304	1	2	2	1

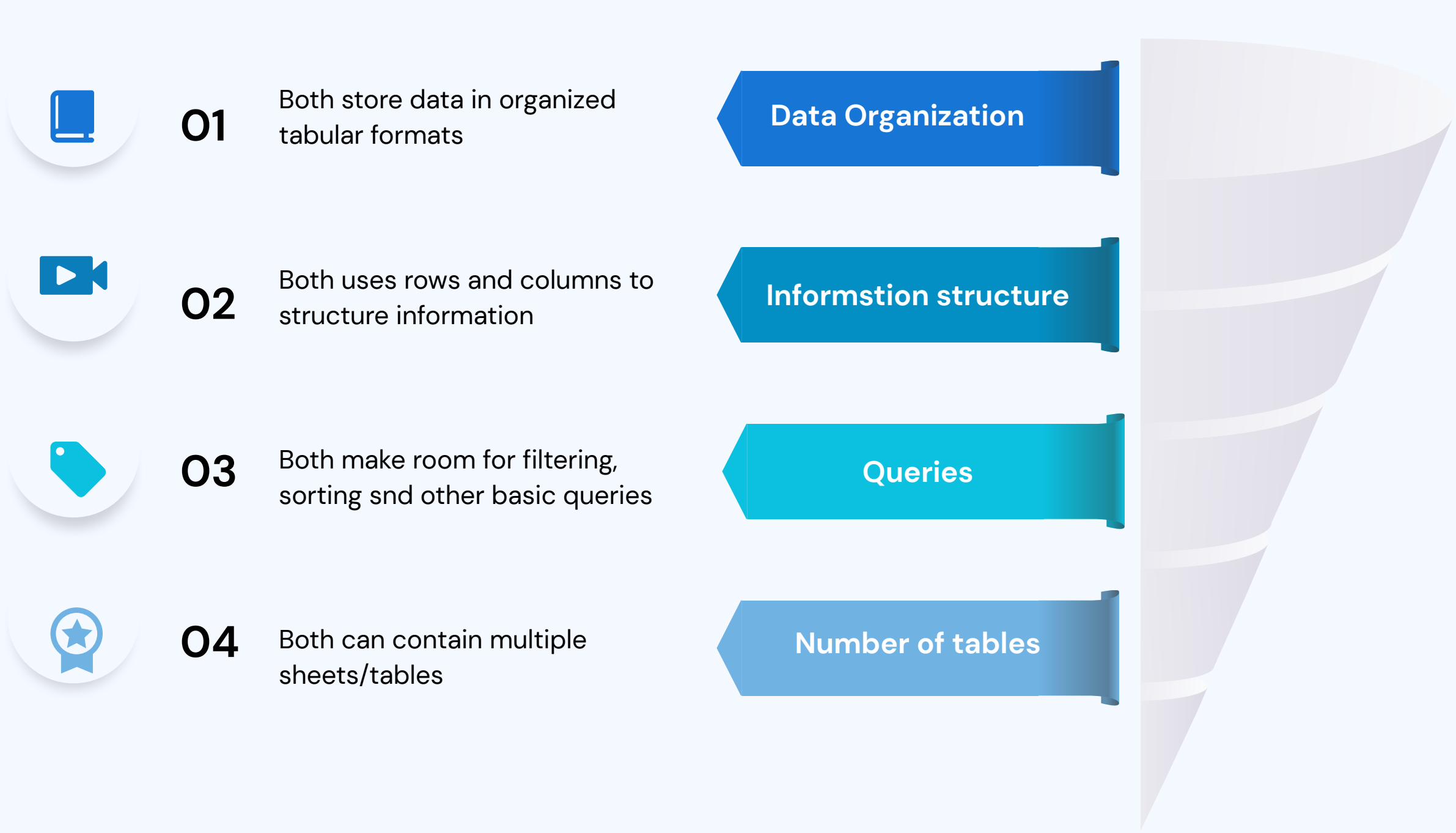
Relational Database

These are databases that come in the form of tables, like an excel spreadsheet. It is laid out in rows and columns. All relational databases use/speak a language known as SQL.

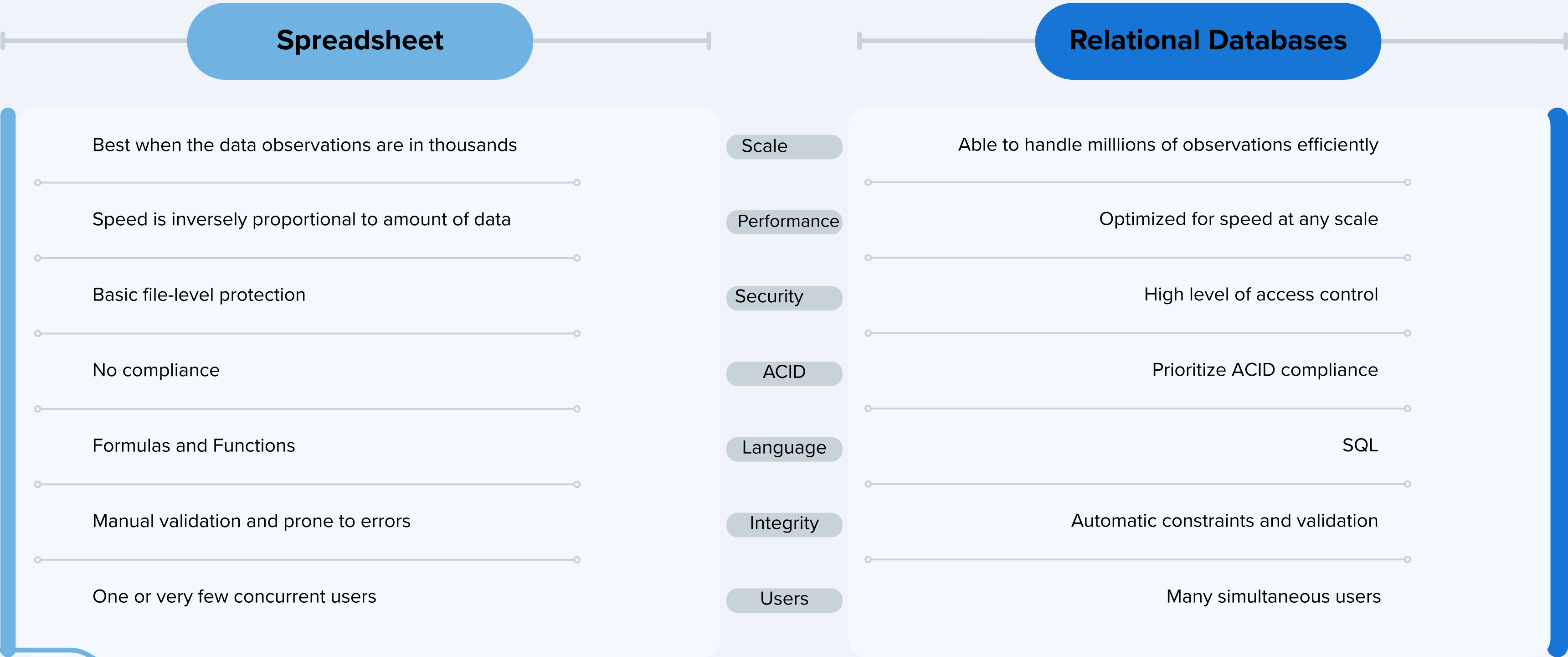


Databases and Spreadsheet Similarities

Tahap Kesadaran



Databases and Spreadsheet Differences



The pros of a Database is coherent in its **Structure**

Structure of database

Structure is how data is arranged and organized in database.

Why Structure Matter

Integrity: Structure products data quality through certain standard rules:

Example:

A student's age cannot be negative

Duplicated email must be prevented

Scalability: Structured database takes into account the possibility of growing from hundreds to trillions of records without degrading in performance.

Consistency: Structure ensures that the data is entered the same way every.

Eg. Phone numbers and dates

233-1234567

DD-MM-YYYY

Database Shema

Shema

This is the overall design or blueprint of your database



01

External/View

This has the highest level of abstraction.
It defines how different users see and interact with the database
Eg. In an online shop website, how the administrator views the site is different from how a visitor views it.



03

Conceptual/Logical

This also has a high level of abstraction.
It defines what data is stored, the data types, constraints, and relationships between tables
Eg. table customer has cust_id(pk) and email (varchar, unique), and it is related to table order

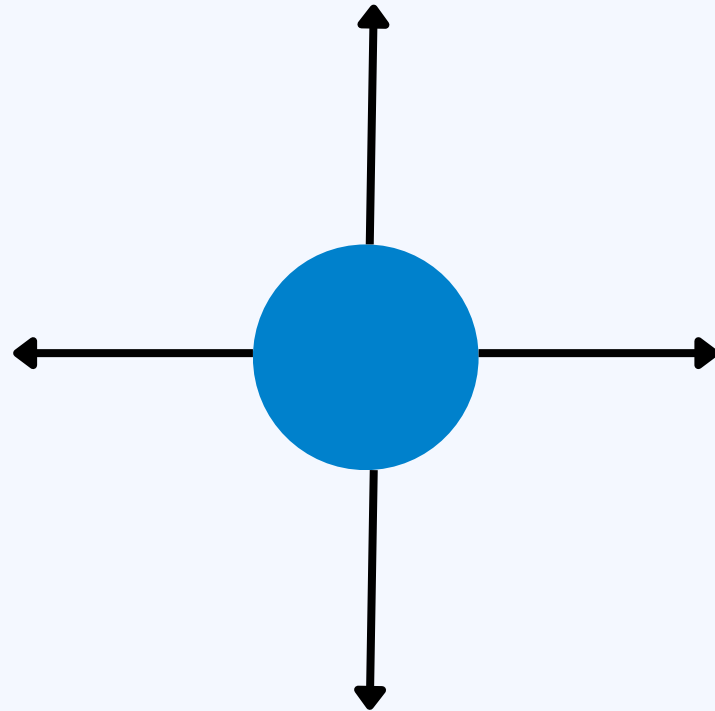


05

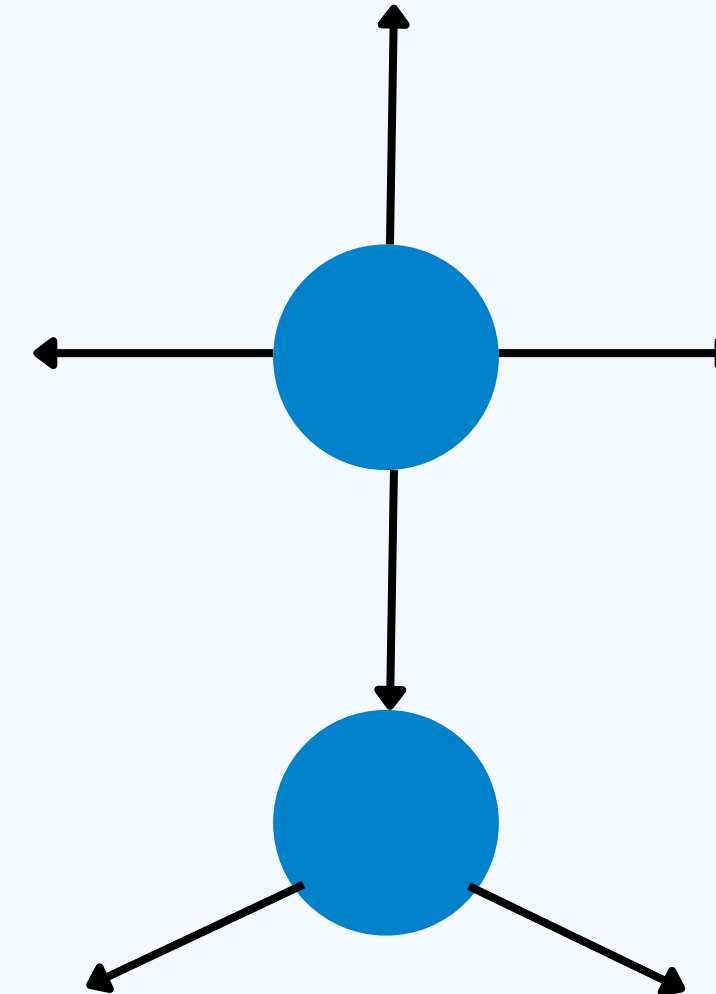
Physical

This has the lowest level of abstraction.
It defines how the data is stored physically and optimized in the system.
Eg. choosing char datatype for storing tel number optimizes space compared to using varchar

Star Schema



Snowflakes



INSTEAD OF JUST THROWING ALL DATA INTO A SINGLE PILE,
STRUCTURE ENSURES THAT DATA IS ORGANIZED INTO

Tables as **Entities**

Rows as **Records**

Columns as **Attributes**



Relational model organizes data into interconnected tables where each table represents entities. So, instead of storing customer and product information in every order row:

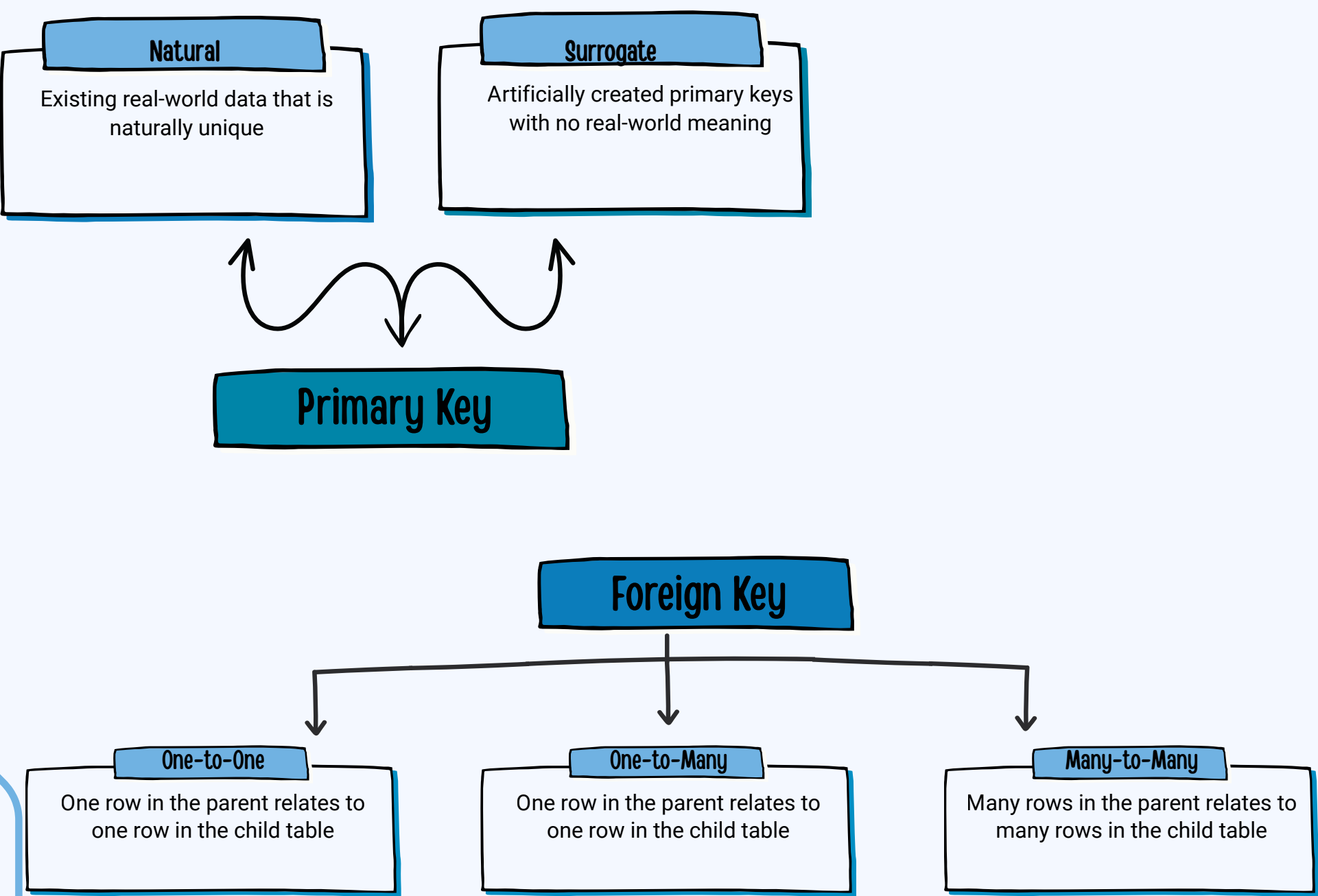
Customers once in '**Customer Table**'

Product once in '**Product Table**'

Then the '**Order Table**' just reference them

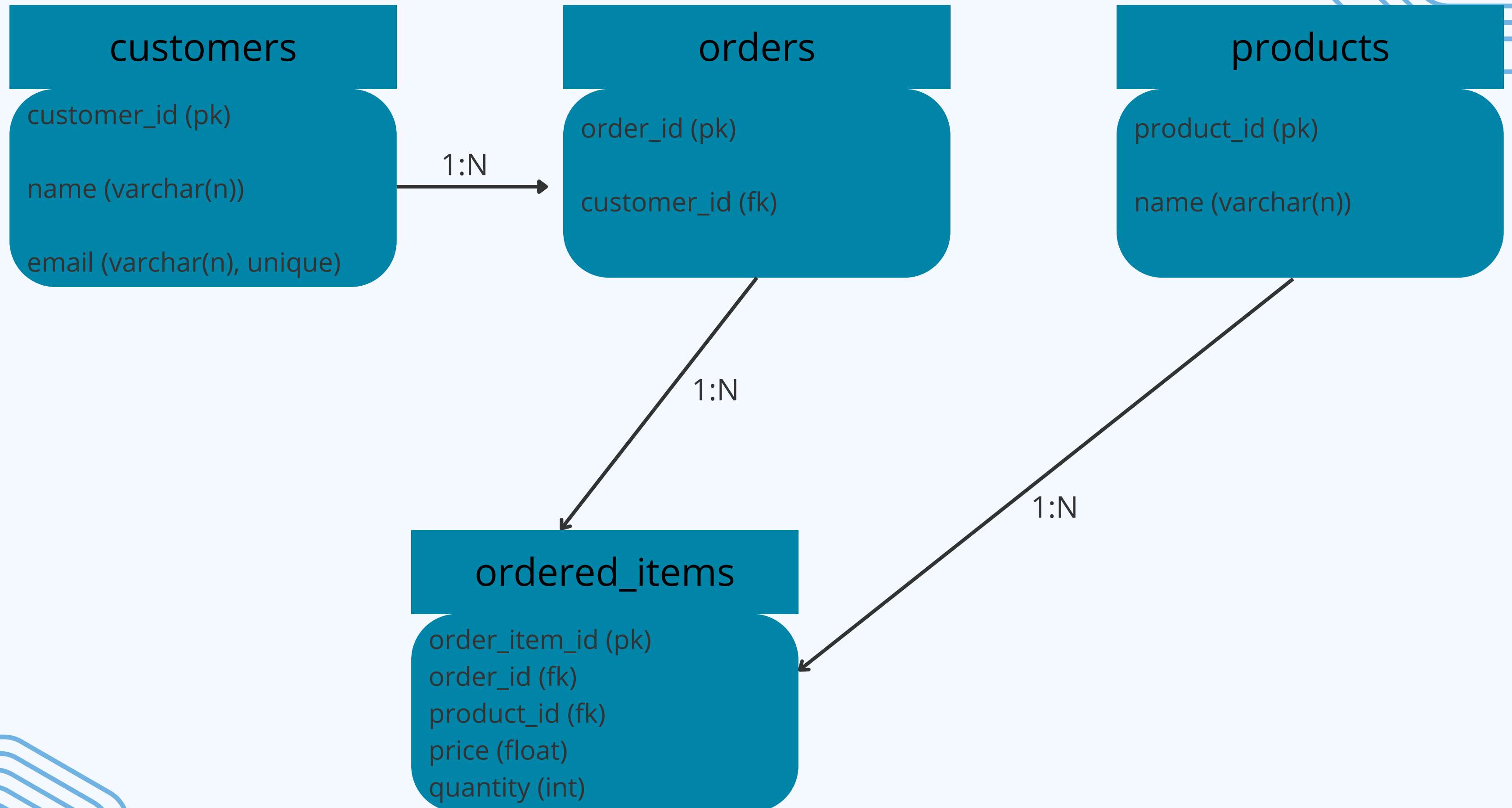
Keys in Database

Special Columns (combination of columns) used to identify rows and link tables together



Primary Key	Main identifier. No two rows can have same key value
Foreign Key	Column(s) that reference the primary key of another table
Candidate Key	Any column (set of columns) that can act as primary key
Composite Key	Primary key made of multiple columns together
Super Key	Any set of columns including the primary key that can uniquely identify a row

Relational Model



Normalization in Database

The process of organizing data to reduce redundancy and improve consistency



Wasting storage due to repeated information (redundancy)

ID	STUDENT_NAME	STUDENT_EMAIL	COURSE_ID	COURSE_NAME	INSTRUCTOR	GRADE
1	Grace Koku	gk@gmail.com	MATH 210	Calculus	Larry Klutse	84
2	Grace Koku	gk@gmail.com	MATH 212	Linear Algebra	Bismarck Duah	91
3	Grace Koku	gk@gmail.com	MATH 214	Abstract Algebra	Juliet Ansah	75
4	Bright Andor	ba@gmail.com	MATH 212	Linear Algebra	Bismarck Duah	88
5	Bright Andor	ba@gmail.com	MATH 210	Calculus	Larry Klutse	90

Future changes lead to many updates (update anomalies)

Inability to introduce new item due to incomplete data (insert anomalies)

Unintended data loss (delete anomalies)

Good or Bad Table?

Data Types

Data types enforces consistency, validation, and optimization by specifying what kind of data can be stored in a column

Common Data Types

- **Text Data**

- VARCHAR(n) - Stores text up to n characters
- TEXT - Stores unlimited amount of text
- CHAR(n) - Store fixed number of characters

- **Numeric Data**

- INTEGER - Any whole number (positive, negative, zero)
- BIGINT - Stores very large whole numbers

- **DECIMAL(p,s):** Stores fixed point decimal, where p is total digit and s is decimal places

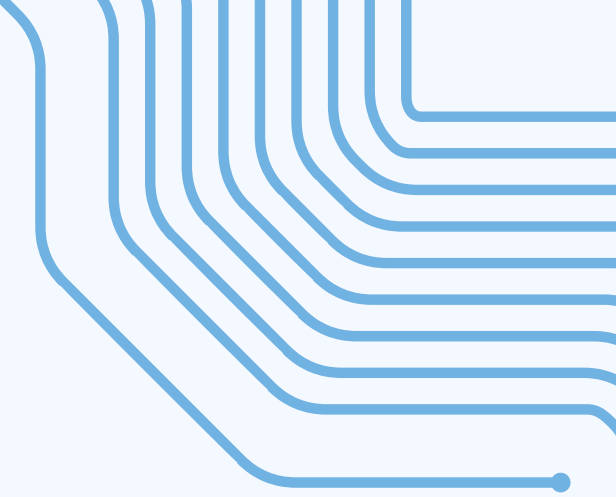
- **FLOAT/DOUBLE:** Stores approximate decimal numbers

- **BOOLEAN** - Store only two values (True/False)

- **Date and Time Data**

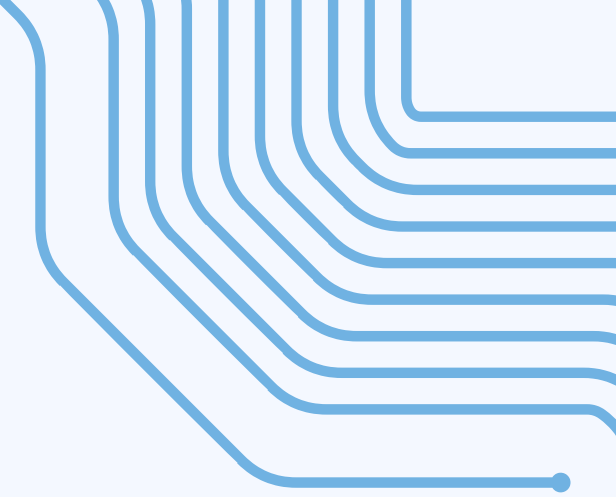
- DATE - Stores only date
- TIME - Stores only time (hour, minute, second)
- TIMESTANP - Stores date and time

Data Constraints



- NOT NULL - The purpose is to ensure that the column must have a value. Example; name VARCHAR (100) NOT NULL
 - UNIQUE - Ensures there are no duplicated values. Example; email VARCHAR (255) UNIQUE
 - DEFAULT - Provides an Auto-set. Example; discount DECIMAL (5,2) DEFAULT 0
 - CHECK - Validate data condition. Example; price DECIMAL (10,2) CHECK (price >0)
- 

Data Administration



Database Administration is the management and maintenance of databases to ensure they are secured, reliable, selectively available, and performant.

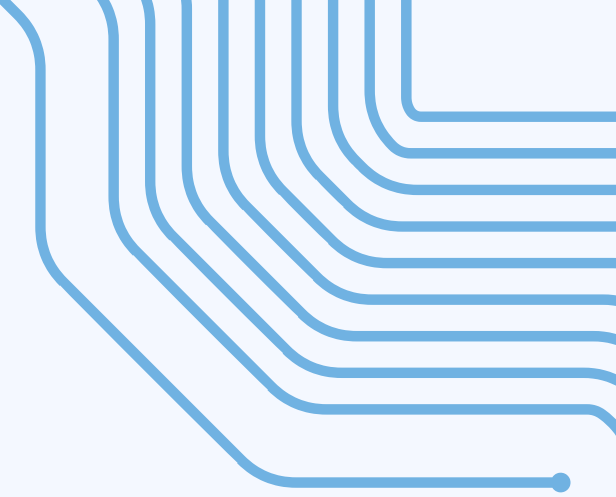
Key Responsibility:

1. Security and Access Control

What it means

- Control who can access the database
 - Control what each user can access and do
 - Protecting sensitive data
- 

Data Administration



2. Backup and Recovery

What it means

- Creating copies of data to protect against loss
- The ability to restore data if something goes wrong

3. Performance Monitoring and Optimization

What it means

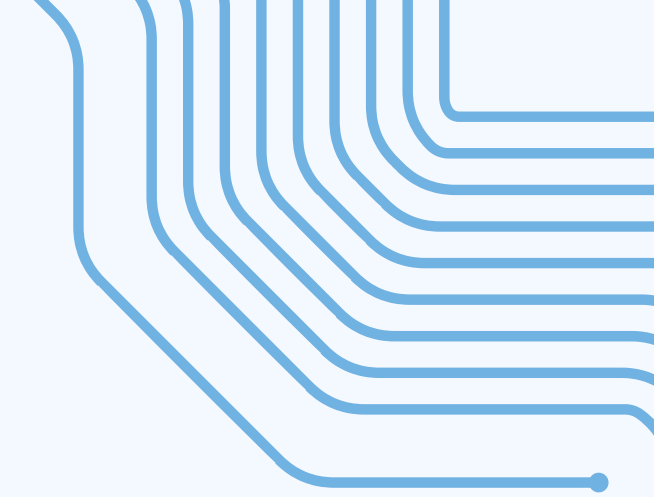
- Ensuring the database runs fast
 - Identifying and fixing slow queries
 - Monitoring resources
- 

4. Maintenance and Updates

What it means

- Keeping the database software current
- Removing unwanted or redundant data

Data Administration



5. Monitoring and Troubleshooting

What it means

- Watching the database for problems
- Fixing problems when they occur
- Responding to emergencies

6. User and Access Management

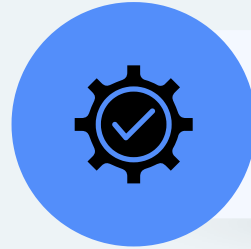
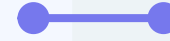
What it means

- Creating and removing user accounts
 - Granting appropriate permissions
 - Managing user credentials
- 

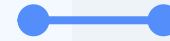
**Database
administration ensures
that databases are**



Secured



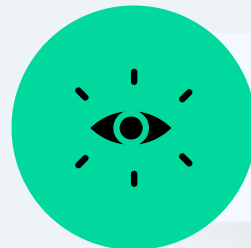
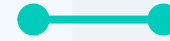
Available



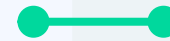
Performant



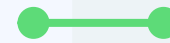
Reliable



Compliant



Maintainable

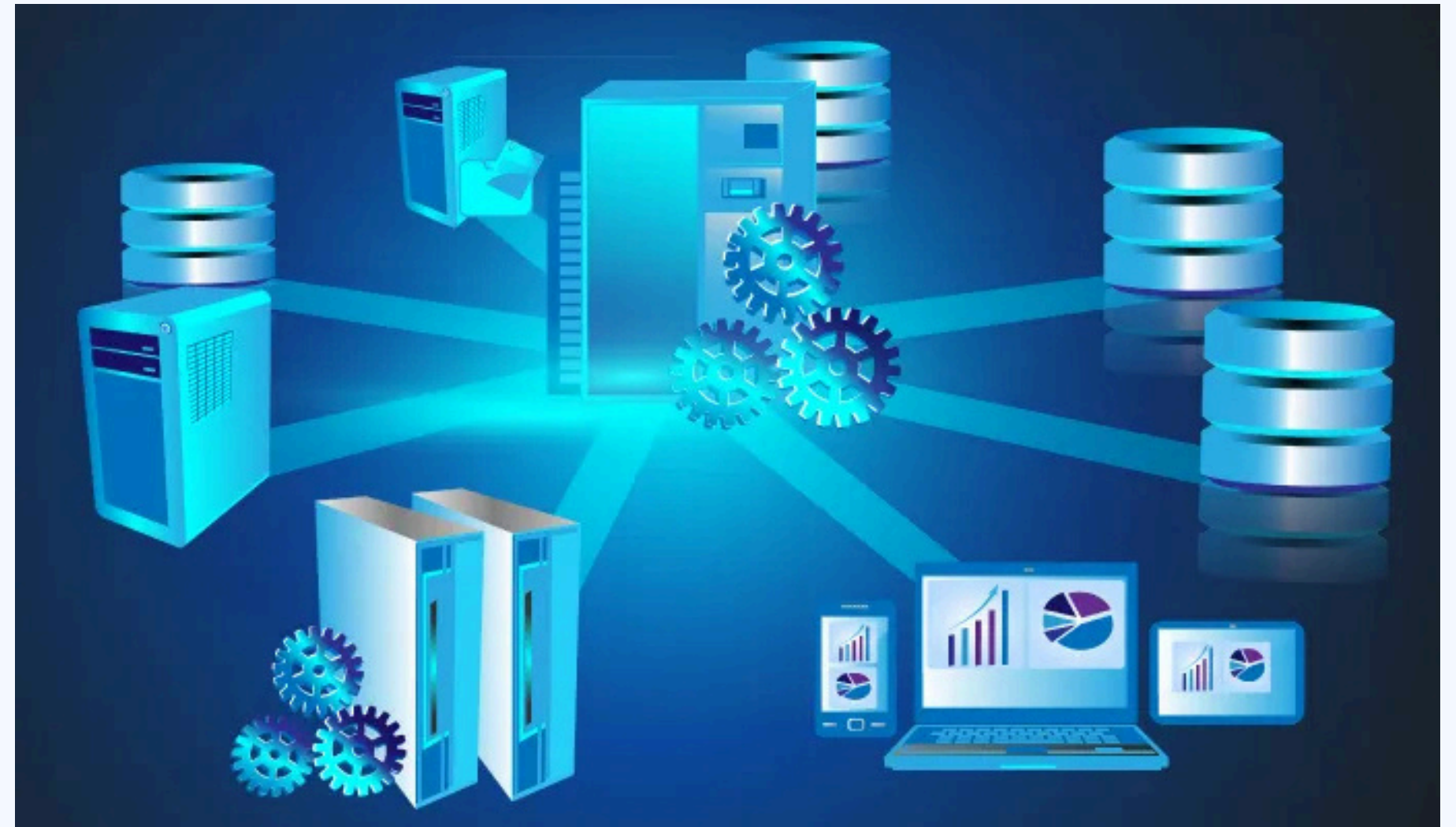


Database management system (dbms)

DBMS is a software that allows users to store, retrieve, manipulate, and manage databases

There are several existing DBMS today:

1. PostgreSQL
2. MySQL
3. SQLite
4. SQL Server
5. Oracle



Introduction to postgResql

PostgreSQL is an open-source relational database management system that is known for its reliability and large-scale application

Why PostgreSQL:

1. Cross-platform
2. Free and open-source
3. Excellent ACID compliance
4. Strong security
5. Support large-scale applications



Getting started with PostgreSQL

<https://www.postgresql.org/download/>

